

PHALABORWA, South Africa

WMO: 681900

Lat: 23.937S

Lon: 31.155E

Elev: 1432

StdP: 13.95

Time Zone: 2.00 (E02)

Period: 98-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	43.9	46.3	21.3	16.8	71.2	26.8	21.9	73.6	16.3	64.3	14.4	64.7	1.4	320	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	98.3	70.4	95.8	70.1	93.4	69.7	79.1	86.8	77.6	85.4	76.2	84.2	5.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	149.2	82.9	75.5	140.7	81.5	73.8	133.1	80.1	43.9	87.0	42.2	85.1	40.7	84.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.2	12.1	39.8	103.7	2.7	2.3	37.8	105.4	36.2	106.7	34.7	108.1	32.8	109.8		
(5)			36.5	78.9	2.5	2.9	34.7	81.0	33.2	82.7	31.8	84.3	30.0	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.1	78.6	78.6	77.2	72.2	67.8	63.9	62.6	66.1	70.9	73.8	76.2	77.6
	DBStd	7.31	4.12	3.55	4.35	4.36	4.66	4.07	3.68	4.66	5.63	5.69	5.77	5.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	241	0	0	0	2	21	69	92	41	11	3	2	0
	CDD50	8065	888	801	842	667	552	416	392	500	628	736	787	856
	CDD65	2832	423	381	378	220	107	35	18	76	189	275	339	391
	CDH74	27441	3918	3330	3213	1689	1140	554	437	1083	2192	2798	3314	3773
	CDH80	11668	1662	1416	1379	598	354	126	93	369	1017	1321	1594	1739
Wind	WSAvg	4.6	5.2	5.1	4.5	4.0	3.2	3.4	3.9	4.4	5.7	5.5	5.1	5.2
Precipitation	PrecAvg	17.4	3.2	3.3	2.3	0.7	0.2	0.2	0.2	0.1	0.2	1.0	2.5	3.7
	PrecMax	39.9	7.6	17.8	9.0	2.6	1.2	1.1	1.4	0.4	0.9	2.2	6.4	7.6
	PrecMin	6.5	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.6
	PrecStd	8.0	2.0	4.2	2.2	0.6	0.3	0.3	0.3	0.1	0.3	0.7	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.1	98.4	98.7	93.9	93.1	88.0	86.5	92.5	98.7	101.0	100.8	98.8
		MCWB	75.6	73.1	72.4	69.2	64.6	59.6	57.6	61.1	63.2	68.0	70.2	72.1
	2%	DB	93.8	94.1	94.5	90.3	87.6	84.0	82.6	88.0	94.3	96.2	96.9	95.8
		MCWB	73.8	72.1	72.3	68.3	63.5	59.2	58.4	60.7	63.6	67.7	69.3	71.9
	5%	DB	91.2	91.1	90.8	87.1	83.8	80.4	79.1	84.5	90.0	92.0	93.1	93.1
		MCWB	73.2	71.5	70.9	68.2	62.5	59.7	58.9	59.5	63.2	67.2	69.4	72.0
	10%	DB	88.2	88.3	87.6	83.4	80.4	77.0	75.4	80.4	86.1	87.8	89.4	89.6
		MCWB	73.2	71.8	70.8	68.1	62.6	59.8	58.6	60.3	64.1	67.1	70.0	72.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	80.8	80.2	77.7	72.8	68.0	66.5	68.1	72.6	75.5	78.0	78.4
		MCDB	87.6	86.7	88.5	88.7	84.3	76.1	74.8	77.9	87.2	88.6	87.7	89.3
	2%	WB	78.9	78.6	78.2	74.5	69.3	65.7	64.5	65.9	69.5	72.2	75.4	76.4
		MCDB	85.3	84.4	86.8	83.3	79.5	74.9	74.4	76.5	83.1	84.8	84.5	86.9
	5%	WB	77.1	76.9	76.3	72.1	67.1	63.9	62.5	64.0	67.2	70.3	73.6	74.9
		MCDB	84.1	83.6	83.4	79.2	77.4	73.3	71.6	75.5	81.1	84.2	83.9	85.4
	10%	WB	75.4	75.3	74.3	70.2	65.1	62.0	60.6	62.4	65.7	68.7	72.2	73.7
		MCDB	83.5	82.6	81.3	78.1	74.6	71.9	70.1	74.2	79.1	82.0	82.8	84.5
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.7	18.5	19.9	25.2	26.7	25.9	25.8	24.9	21.4	19.8	19.0
		MCDBR	22.4	23.2	24.7	27.3	31.4	32.6	33.5	34.1	33.9	31.3	27.8	25.3
		MCWBR	6.4	6.5	7.3	10.2	11.4	13.5	13.9	13.2	11.7	9.7	8.2	7.4
	5% WB	MCDBR	18.1	16.5	18.4	21.1	24.0	25.8	26.5	27.3	26.6	24.1	21.1	20.5
		MCWBR	8.4	9.2	10.1	10.3	11.6	14.0	15.2	13.9	11.6	9.1	8.7	7.3
Clear-Sky Solar Irradiance	taub	0.410	0.402	0.396	0.381	0.351	0.335	0.337	0.391	0.453	0.463	0.429	0.423	
	taud	2.376	2.401	2.408	2.406	2.439	2.459	2.444	2.283	2.139	2.166	2.297	2.335	
	Ebn at Noon	297	295	287	277	273	272	275	270	266	274	289	293	
	Edh at Noon	42	40	38	35	31	29	30	39	49	49	45	43	
All-Sky Solar Radiation	RadAvg	1943	1897	1694	1451	1337	1195	1251	1467	1658	1737	1840	1911	
	RadStd	135	157	128	93	60	76	48	86	113	109	161	176	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page